

VIDIN, Bulgaria

WMO: 155020

Lat: 43.983N

Lon: 22.850E

Elev: 102

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	14.9	5.0	7.1	9.0	11.4	9.9	15.9	20.3	38.3	15.9	39.0	1.8	320	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	93.9	73.2	91.0	72.3	88.2	71.5	75.5	89.2	74.0	87.5	72.6	85.6	4.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	115.2	83.3	69.6	109.1	81.1	68.1	103.7	79.3	39.0	89.4	37.5	87.7	36.2	85.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	12.1	10.5	1.3	98.5	7.8	3.8	-4.3	101.2	-8.8	103.5	-13.2	105.6	-18.9	108.4		
			0.8	79.3	7.7	2.0	-4.8	80.7	-9.3	81.9	-13.7	83.0	-19.3	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	31.7	35.9	45.3	54.6	63.5	70.5	74.4	73.7	64.9	53.6	43.4	33.8
	DBStd	16.60	9.31	9.86	8.21	6.13	5.57	5.37	4.51	4.81	5.97	7.17	7.21	8.17
	HDD50	1954	570	399	188	26	0	0	0	0	0	46	221	504
	HDD65	4938	1033	813	612	313	96	14	1	4	76	360	649	967
	CDD50	3367	1	6	41	165	418	616	756	736	447	156	23	2
	CDD65	874	0	0	0	3	48	179	292	275	72	5	0	0
	CDH74	9691	0	0	8	79	609	1977	3220	2963	767	67	1	0
	CDH80	3914	0	0	0	13	127	757	1442	1338	228	9	0	0
Wind	WSAvg	3.5	3.4	4.1	4.8	4.1	3.5	3.5	3.3	3.0	3.0	3.0	3.1	3.3
Precipitation	PrecAvg	22.1	1.5	1.4	1.5	1.9	2.2	2.1	2.1	1.7	1.9	2.0	1.7	2.0
	PrecMax	38.8	3.0	3.3	3.5	5.1	4.5	5.1	4.9	6.7	5.6	5.9	4.1	5.2
	PrecMin	10.9	0.3	0.0	0.2	0.0	0.9	0.4	0.2	0.0	0.1	0.2	0.0	0.0
	PrecStd	5.8	0.8	1.0	1.0	1.0	1.0	1.2	1.4	1.5	1.4	1.5	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	65.7	73.8	81.9	87.1	95.2	99.2	98.0	90.7	81.0	69.4	59.8
		MCWB	49.5	53.3	56.7	63.2	69.7	73.2	72.9	74.0	69.7	65.3	58.2	50.5
	2%	DB	52.9	60.1	68.6	75.6	83.5	91.0	94.0	93.7	85.9	74.8	62.1	51.9
		MCWB	45.7	50.2	54.4	60.6	67.6	72.8	73.2	73.1	68.0	62.9	54.5	45.9
	5%	DB	47.8	54.7	64.2	71.3	80.6	87.7	90.8	90.8	82.3	70.4	57.5	47.4
		MCWB	42.9	47.1	52.2	58.5	66.1	71.9	72.5	72.2	66.9	60.6	52.8	43.1
	10%	DB	43.3	49.6	59.6	67.9	77.3	84.5	87.8	87.8	78.5	66.1	54.0	44.2
		MCWB	40.1	43.9	49.9	56.8	64.9	70.5	71.6	71.2	65.5	58.3	50.9	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	54.0	58.2	65.0	71.6	76.8	77.1	78.0	72.0	67.3	59.7	51.1
		MCDB	58.4	64.5	71.1	77.5	84.2	89.9	91.3	91.2	85.7	77.9	66.8	58.7
	2%	WB	46.1	50.4	55.3	62.1	69.1	74.6	75.3	75.5	69.4	64.1	55.9	46.8
		MCDB	52.2	58.9	66.5	73.5	80.8	87.6	89.4	89.0	82.0	72.6	60.4	51.0
	5%	WB	43.1	47.5	53.0	59.7	67.3	72.9	73.8	73.7	67.7	61.4	53.6	43.8
		MCDB	46.9	53.6	63.0	69.4	78.3	85.3	87.7	87.2	79.3	69.1	56.4	46.6
	10%	WB	40.5	44.6	50.5	57.5	65.5	71.1	72.4	72.0	66.2	59.0	51.2	41.6
		MCDB	43.3	49.1	58.6	66.0	75.6	82.6	85.4	85.2	76.6	65.1	53.9	43.8
Mean Daily Temperature Range	5% DB	MDBR	12.4	15.3	19.6	21.3	21.9	22.8	24.1	24.9	22.0	18.4	12.9	12.4
		MCDBR	20.8	25.2	29.1	28.6	28.4	29.1	30.2	30.9	29.5	26.9	21.8	19.1
		MCWBR	14.3	16.8	16.9	15.6	14.2	13.1	12.3	12.8	13.8	15.7	14.5	13.4
	5% WB	MCDBR	19.1	23.6	26.7	25.8	25.9	26.7	26.8	27.3	25.9	24.6	17.7	17.4
MCWBR		13.4	16.0	16.2	14.6	13.8	12.8	12.0	12.7	13.0	15.0	12.2	12.8	
Clear-Sky Solar Irradiance	taub	0.347	0.375	0.414	0.476	0.456	0.450	0.461	0.479	0.448	0.405	0.373	0.346	
	taud	2.285	2.199	2.165	2.016	2.138	2.198	2.169	2.115	2.165	2.276	2.313	2.310	
	Ebn at Noon	239	252	258	251	260	262	257	247	244	239	227	226	
	Edh at Noon	27	35	42	52	48	45	46	47	41	32	27	24	
All-Sky Solar Radiation	RadAvg	446	704	1073	1448	1765	1990	2008	1789	1280	836	485	376	
	RadStd	65	102	138	166	156	167	135	157	110	118	68	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+1.06	N/A	N/A	N/A	+0.68	N/A	N/A	-268	+167	N/A

Nomenclature: See separate page